Less repeated work

Why do different approaches lead to the same result?

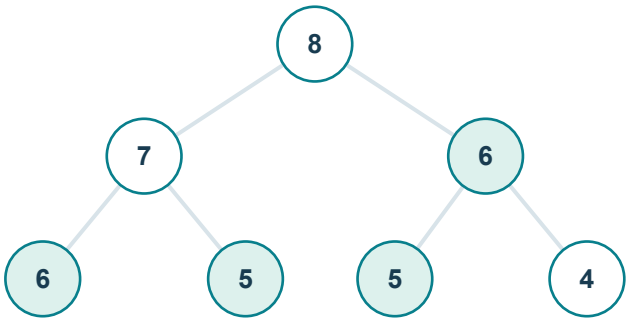
Each new number is the sum of the previous two. See how often the same subproblem appears and how to avoid repeating the work.

Explore a number sequence DRAFT

Target: number 8 | Compare approaches | Single step

Result: 21

0, 1, 1, 2, 3, 5, 8, 13, 21



Each circle represents a subproblem.

Already calculated?

Subproblems 6 and 5 appear more than once. Stored answers can be reused.

Same result, different approaches.

EXPLORE IT STEP BY STEP

1. Choose a target
2. Follow the steps
3. Spot repeated work

WHAT YOU WILL DISCOVER

Discover why storing an intermediate result can save a great deal of repeated work.

The diagram shows calculation steps, not measured speeds.

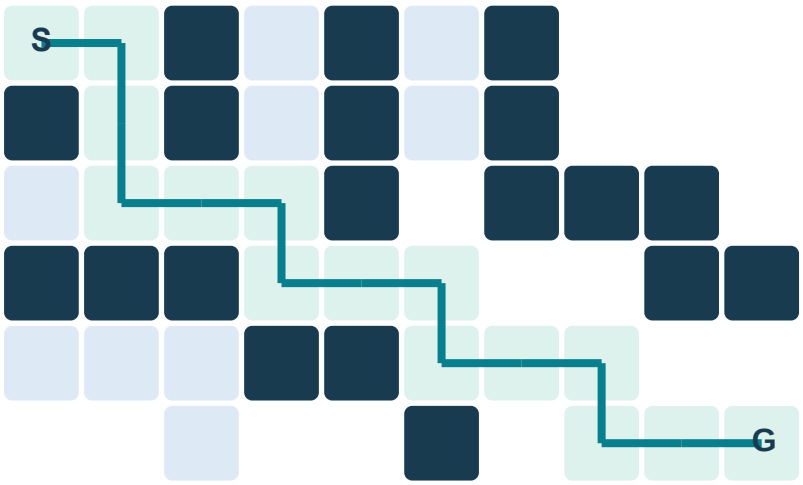
Find a way through the maze

Which search reaches the goal with fewer detours?

Draw walls, set the start and goal, and watch different search methods explore the maze.

Compare three search methods DRAFT

Draw walls | Start search | Single step



Found path

Explored cells

Walls

Start S / Goal G

Teal: found path. Light blue: cells already explored.

EXPLORE IT STEP BY STEP

1. Design a maze
2. Watch the search
3. Compare paths

WHAT YOU WILL DISCOVER

Distinguish a path from the shortest path and see how guidance towards the goal changes the search.

Every step has the same cost in this example.

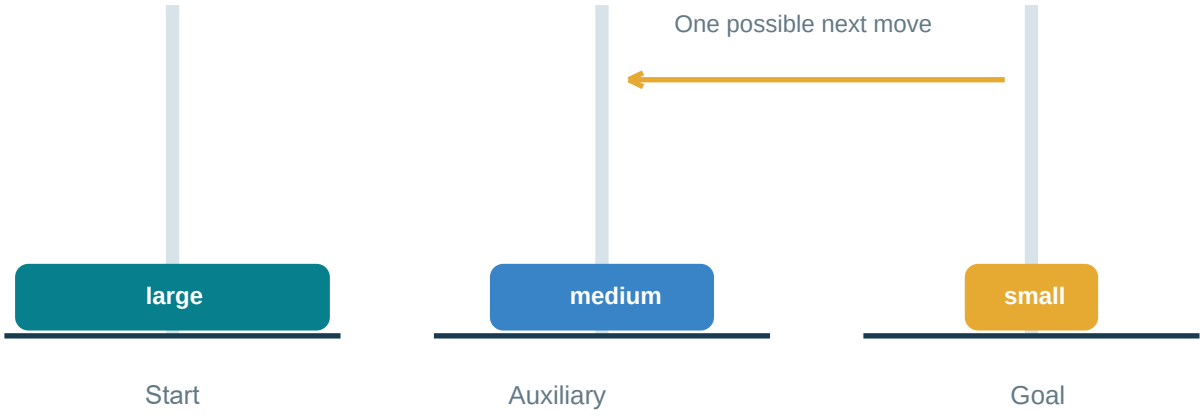
Move the Towers of Hanoi

How can all disks reach the right-hand tower?

Move only one disk at a time. A larger disk must never sit on a smaller one. Try it yourself or reveal the next move.

Three disks, one clear goal DRAFT

3 disks | Next move | Restart



Example state after 2 moves. With 3 disks, a total of 7 moves is enough.

EXPLORE IT STEP BY STEP

1. Choose disks
2. Make legal moves
3. Understand substeps

WHAT YOU WILL DISCOVER

See how a large rearrangement problem consists of smaller problems of the same kind.

Adding disks quickly increases the number of moves.

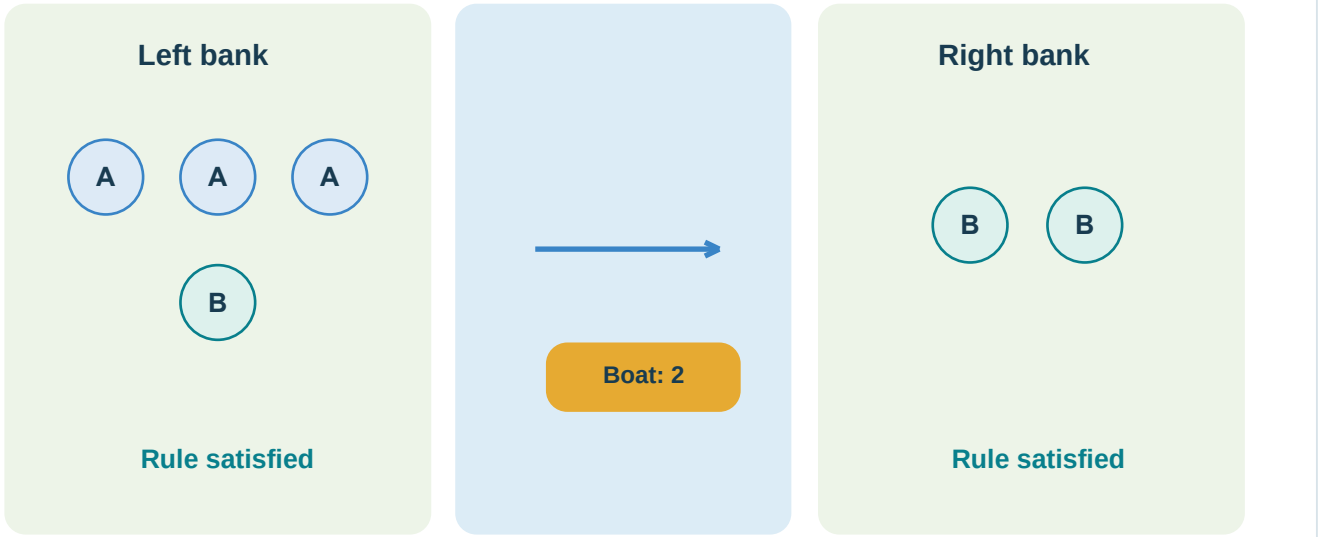
Get everyone safely across

Which crossings bring everyone to the other bank?

The boat holds at most two people. Wherever group A is present, group B must not outnumber it. Find a legal sequence of crossings.

Two groups and a small boat DRAFT

Choose people | Check crossing | Show solution



Example: 3 from A and 1 from B on the left; 2 from B on the right. Both banks are valid.

EXPLORE IT STEP BY STEP

1. Choose passengers
2. Check both banks
3. Find a move sequence

WHAT YOU WILL DISCOVER

Discover why a good next move must also allow later return crossings.

Illustration of the classic Missionaries and Cannibals puzzle.

Satisfy every rule at once

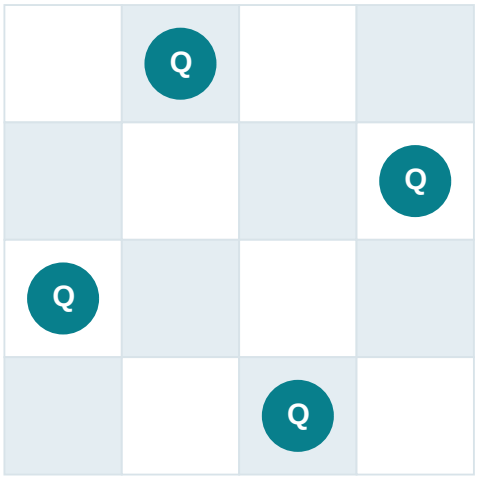
How do you find a solution that meets every rule?

Place queens on the board so that none attacks another. At a dead end, undo a decision and try another option.

A small board full of constraints

DRAFT

Queens | Map | Words | Solve | Single step



One possible solution, not the only one.

All constraints satisfied

- One queen per row
- One queen per column
- No shared diagonal

Example: 4 queens on a 4 x 4 board, with no queen attacking another.

EXPLORE IT STEP BY STEP

1. Set the rules → 2. Check options → 3. Backtrack from dead ends

WHAT YOU WILL DISCOVER

The same principle helps with map colouring, word grids and cryptarithmetic puzzles such as SEND + MORE = MONEY.

The prototype shows queens; use the selector to switch to other puzzles.

A schedule without double bookings

What happens when two people want the same room?

Request a suitable schedule. If someone books the room in the meantime, the application shows the conflict and helps you find an alternative.

Check a proposal and confirm a booking DRAFT

Suggest a schedule | Check booking | Choose alternative

Time	Room A	Room B
09:00	Book club	
10:00	Now occupied	Available alternative
11:00		Music

Request: Tuesday, 10:00, an available room

Room A at 10:00 has since been booked. Room B is still available as an alternative.

EXPLORE IT STEP BY STEP

1. Enter preferences
2. Get a proposal
3. Check current status

WHAT YOU WILL DISCOVER

Understand the difference between a suitable proposal and an actually confirmed booking.

All names, rooms and dates are fictional examples.

Read DNA and store it compactly

Where does a particular DNA word occur?

DNA is shown as a sequence of A, C, G and T. Search for a short word and discover how to represent the same sequence more compactly.

A sequence of four letters

DRAFT

Word: ACG | Show matches | Compact view

DNA sequence

A	C	G	T	A	C	G	T
1	2	3	4	5	6	7	8

Search word

A C G

The same information, compactly

A = 00 C = 01 G = 10 T = 11

Four letters, four short pairs of bits.

ACG appears twice: starting at position 1 and position 5.

EXPLORE IT STEP BY STEP

1. Select a sequence
2. Find a short word
3. Compare representations

WHAT YOU WILL DISCOVER

See search matches and discover that a more compact representation can contain the same information.

Positions displayed in this view are counted from 1.

Getting closer to pi

How can a calculation get closer and closer to pi?

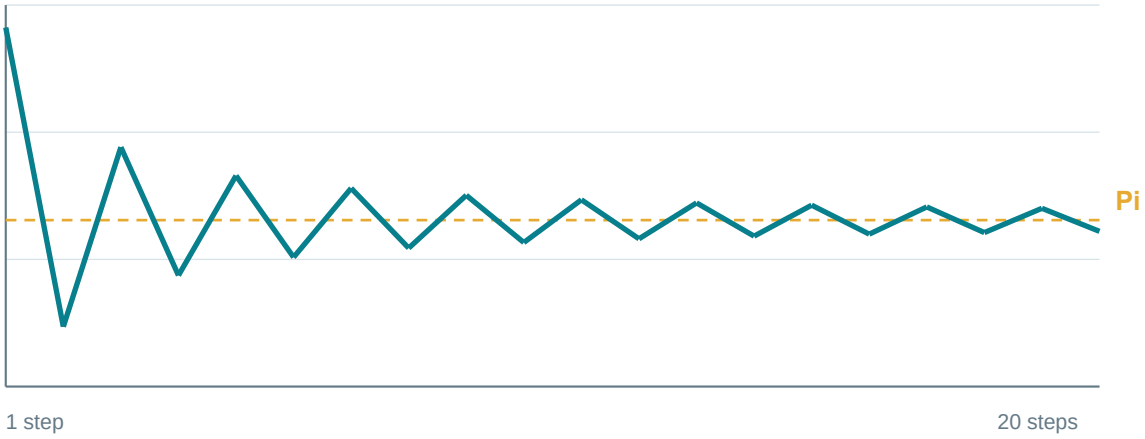
Pi is the ratio of a circle's circumference to its diameter. Watch a calculation whose results alternate above and below the target value.

A new approximation at every step

DRAFT

Steps: 20 | Continue | Show error

Above, then below - the error gets smaller.
Approximation



The curve oscillates around pi. More steps gradually reduce the error.

EXPLORE IT STEP BY STEP

1. Choose step count
2. Watch convergence
3. Compare errors

WHAT YOU WILL DISCOVER

Discover why an approximation may need many steps even when the underlying idea is simple.

The curve illustrates the trend; it is not a live calculation.

A message and a key

How does a key transform a message?

A short message is combined with a key of the same length. Applying that same key again reveals the original message.

Hide a message and reveal it again

DRAFT

Choose example | Encrypt | Decrypt

Message		1	0	1	0
Key	+	0	1	1	0
Hidden message		1	1	0	0

With the same key, 1100 becomes 1010 again.

Example: 1010 combined with 0110 gives 1100; applying 0110 again gives 1010.

EXPLORE IT STEP BY STEP

1. Choose a message
2. Apply the key
3. Recover the message

WHAT YOU WILL DISCOVER

Follow each bit as it changes and see why the correct key matters.

Perfect secrecy in the book's method requires a secret, truly random key that is used only once.

Find the right route

Is the route with the fewest stops also the fastest?

Connect places and assign travel times. Compare a route with few legs against one with the smallest total travel time.

Fewer legs or less total travel time? DRAFT

Start: A / Goal: D | **Legs | Travel time** | Compare

Compare: direct or via two intermediate places?



Teal: 3 + 2 + 4 = 9 minutes

Example: 12 minutes directly; 9 minutes in total via B and C.

EXPLORE IT STEP BY STEP

1. Connect places
2. Set a goal
3. Compare routes

WHAT YOU WILL DISCOVER

Discover that the best route depends on what you want to minimise.

All travel times are fictional example values.

Connect every place at low cost

Which connections do you really need?

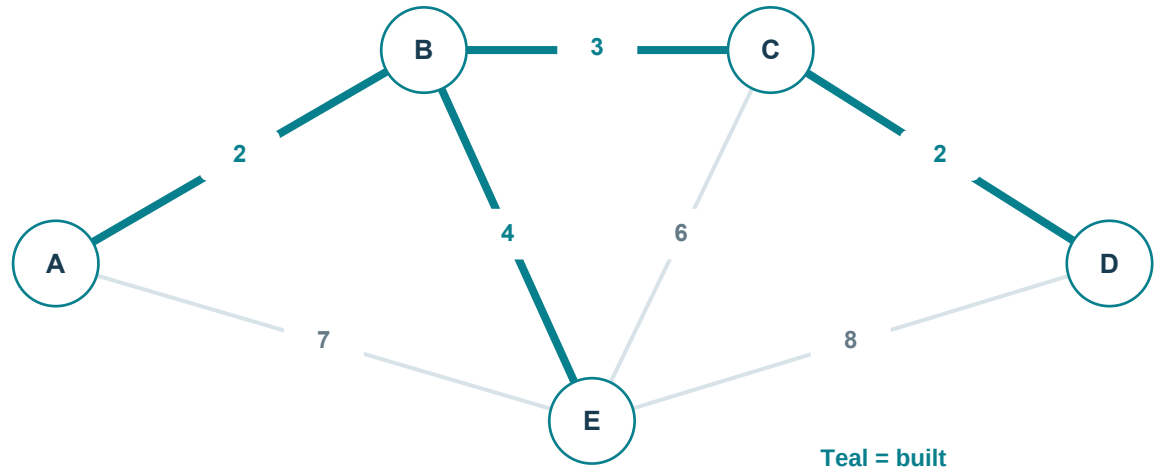
Several places must be reachable from one another. Every possible connection costs money. Choose a connected network with the lowest total cost.

A network with minimal construction cost

DRAFT

Place locations | Change costs | Compare networks

Selected: 2 + 3 + 2 + 4 = 11 cost units



This example connects all 5 places using 4 selected connections.

EXPLORE IT STEP BY STEP

1. Assess connections
2. Build a low-cost network
3. Check total cost

WHAT YOU WILL DISCOVER

See why an extra connection is not always necessary to reach every place.

This model considers construction cost; resilience would be a separate objective.

A game AI thinks ahead

How does a computer choose a good next move?

Play Tic-Tac-Toe or Connect Four. Explore the opponent's possible replies and see which options need no further investigation.

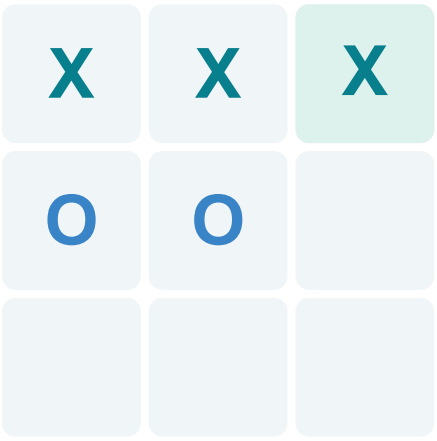
Moves and possible replies

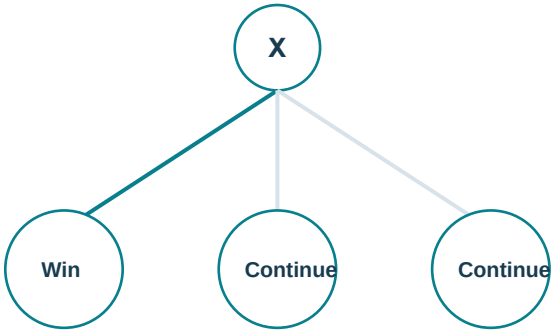
DRAFT

Tic-Tac-Toe | Connect Four

AI move

Show reasoning





An immediate win needs no further replies.

Example: X can win immediately by taking the top-right square.

EXPLORE IT STEP BY STEP

1. Try a move
2. Explore replies
3. Understand the choice

WHAT YOU WILL DISCOVER

Distinguish a move that looks good now from one that also accounts for strong opposition.

The screen shows Tic-Tac-Toe; Connect Four is another view.

What fits in the knapsack?

What do you take when you cannot fit everything in?

Each item has a weight and a value. Choose a set of items and compare it with the best combination within the weight limit.

Limited weight, maximum value

DRAFT

Limit: 7 kg | Check selection | Best selection



Selected: A + B

7 kg / 11 value points

Example: A and B weigh 7 kg together and provide 11 value points.

EXPLORE IT STEP BY STEP

1. Value the items
2. Set a weight limit
3. Compare combinations

WHAT YOU WILL DISCOVER

The most valuable single item does not automatically produce the best overall selection.

Weights and values are simplified examples.

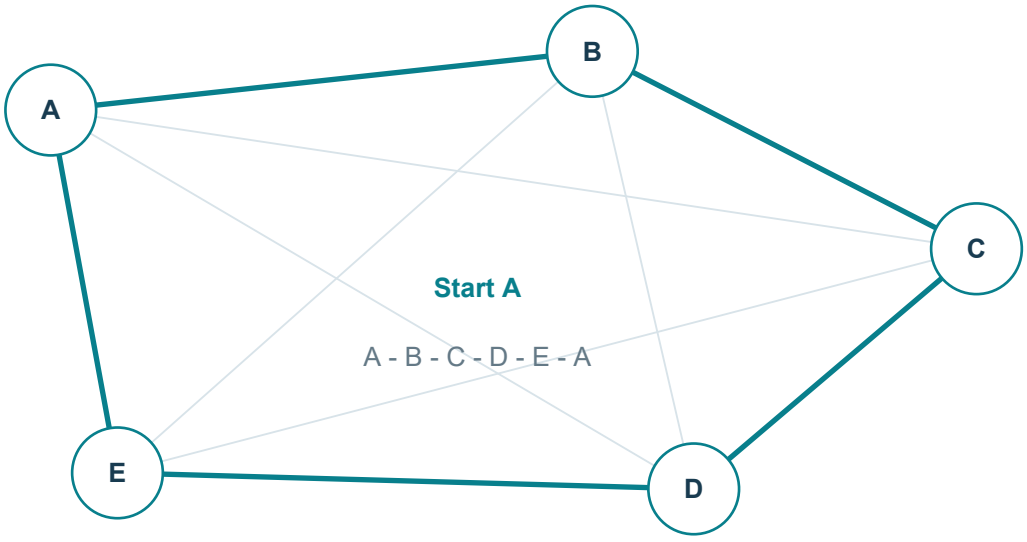
Plan a short round trip

Which visiting order makes a round trip short?

Place a few destinations. The application compares orders that visit each place once and return to the starting point.

Visit every place and return to the start DRAFT

5 places | **Change order** | Compare tours



The coloured tour shows one possible order through all 5 places.

EXPLORE IT STEP BY STEP

1. Place destinations →
2. Compare orders →
3. Find a short tour

WHAT YOU WILL DISCOVER

See how quickly the number of possible orders grows as more places are added.

The sketch illustrates the problem; the drawn route is not a measured travel itinerary.

Words in a phone number

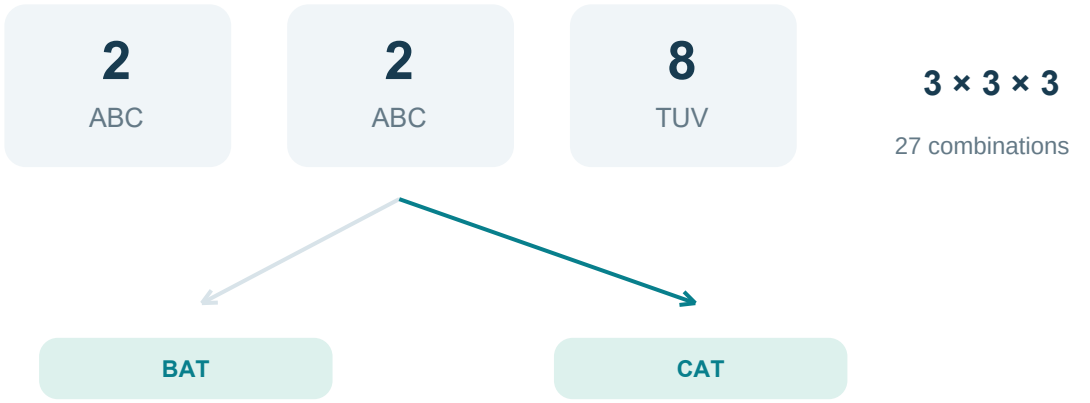
Which words are hidden in a number?

On a phone keypad, several letters share a digit. Try combinations and highlight matching words.

Turn digits into letter combinations

DRAFT

Number: 228 | Show combinations | Words only



A dictionary highlights readable words.

For example, 228 can produce CAT or BAT.

EXPLORE IT STEP BY STEP

1. Enter a number
2. Combine letters
3. Discover words

WHAT YOU WILL DISCOVER

Discover how a few choices can generate many combinations.

The word examples are English; the digit 1 has no letters here.

Improve solutions over generations

Can trial and selection improve a solution?

Many candidate solutions compete. Good candidates are combined and slightly modified. Watch their quality develop over several rounds.

Good ideas produce new candidates

DRAFT

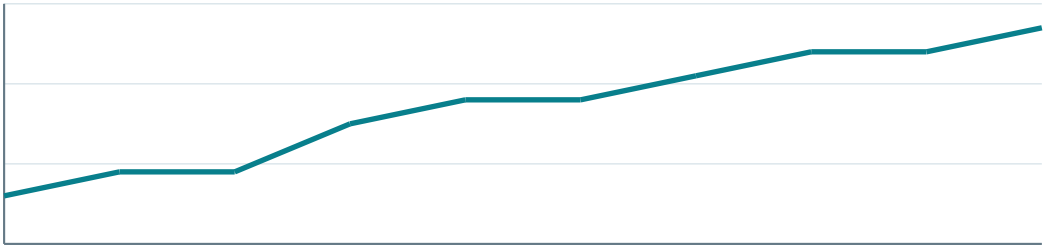
Choose a task | New generation | Multiple starts



Several candidates

Select + modify

Best quality found



Generation 1 — Generation 10

The example curve shows the best solution found so far.

EXPLORE IT STEP BY STEP

1. Generate candidates
2. Select good variants
3. Test new variants

WHAT YOU WILL DISCOVER

See improvement without a guarantee of finding the very best solution. Different starts can produce different results.

Possible tasks: cryptarithmetic puzzles, a numerical function or compact arrangement of a list.

Group similar points

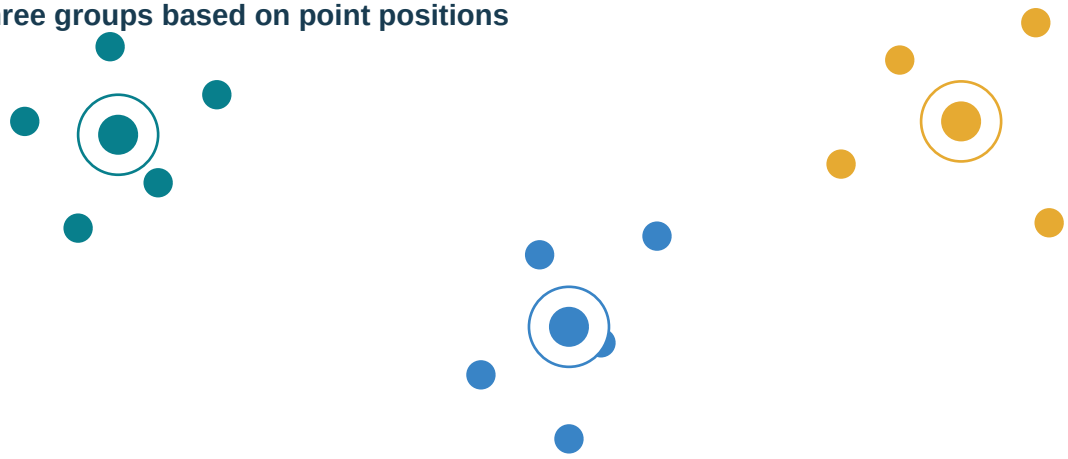
Which data points belong together based on similarity?

Place points or load an example. Choose the number of groups and watch the centres and assignments change.

Individual points become groups DRAFT

Groups: 3 | Reassign | Change start

Three groups based on point positions



Different starting points can produce different groups.

Colours indicate current groups; large circles mark their centres.

EXPLORE IT STEP BY STEP

1. Inspect points
2. Form groups
3. Compare starting points

WHAT YOU WILL DISCOVER

Discover how the number of groups, scale and starting points influence the result.

A computed group gains meaning only through the data it represents.

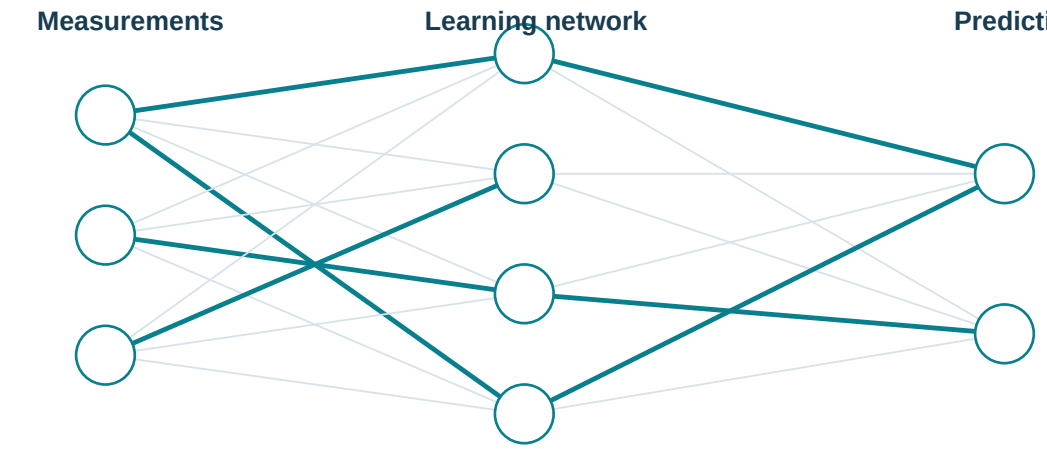
A small network learns from examples

How does a learning model improve its answers?

A small network trains on familiar examples, such as flower measurements. Then test how well it classifies new examples it has not seen before.

Learn to classify from examples DRAFT

Choose sample data | Start training | Test new examples



Train on known examples → **Test on new cases**

Left: measurements. Centre: adjustable connections. Right: predicted class.

EXPLORE IT STEP BY STEP

1. Training examples → 2. Adjust connections → 3. Test on new cases

WHAT YOU WILL DISCOVER

Good answers on familiar examples are not enough. Performance on new cases matters too.

The connections and learning curves are illustrative examples.

Search and understand data freshness

Why do changes sometimes appear later in search?

Search a small document catalogue. Edit an entry and watch when search begins to show its updated version.

Search results may briefly lag behind DRAFT

Search: forest | Edit entry | Refresh search view

forest

A walk in the forest

Search view: previous

A walk in the city park

Source: updated

The search view must first pick up the change.

The source has changed. The search view still shows the previous version.

EXPLORE IT STEP BY STEP

1. Edit a document
2. Observe the delay
3. See updated results

WHAT YOU WILL DISCOVER

Understand the difference between source data and a search view prepared from it.

Pausing updates is part of this experiment.

Where might DNA replication begin?

Which short words and unusual regions stand out?

Explore a DNA sequence using a sliding window. Frequent short words and a plot of base distribution help narrow down interesting regions.

Several clues to a candidate region DRAFT

Move window | Word length: 3 | Similar words

G A C C G A T C G A C C G

Yellow window: a region examined more closely

Distribution of G and C along the sequence



Sequence start — Sequence end

Highlighted region: a candidate that needs checking against further evidence.

EXPLORE IT STEP BY STEP

1. Survey the DNA
2. Find a notable region
3. Check short words

WHAT YOU WILL DISCOVER

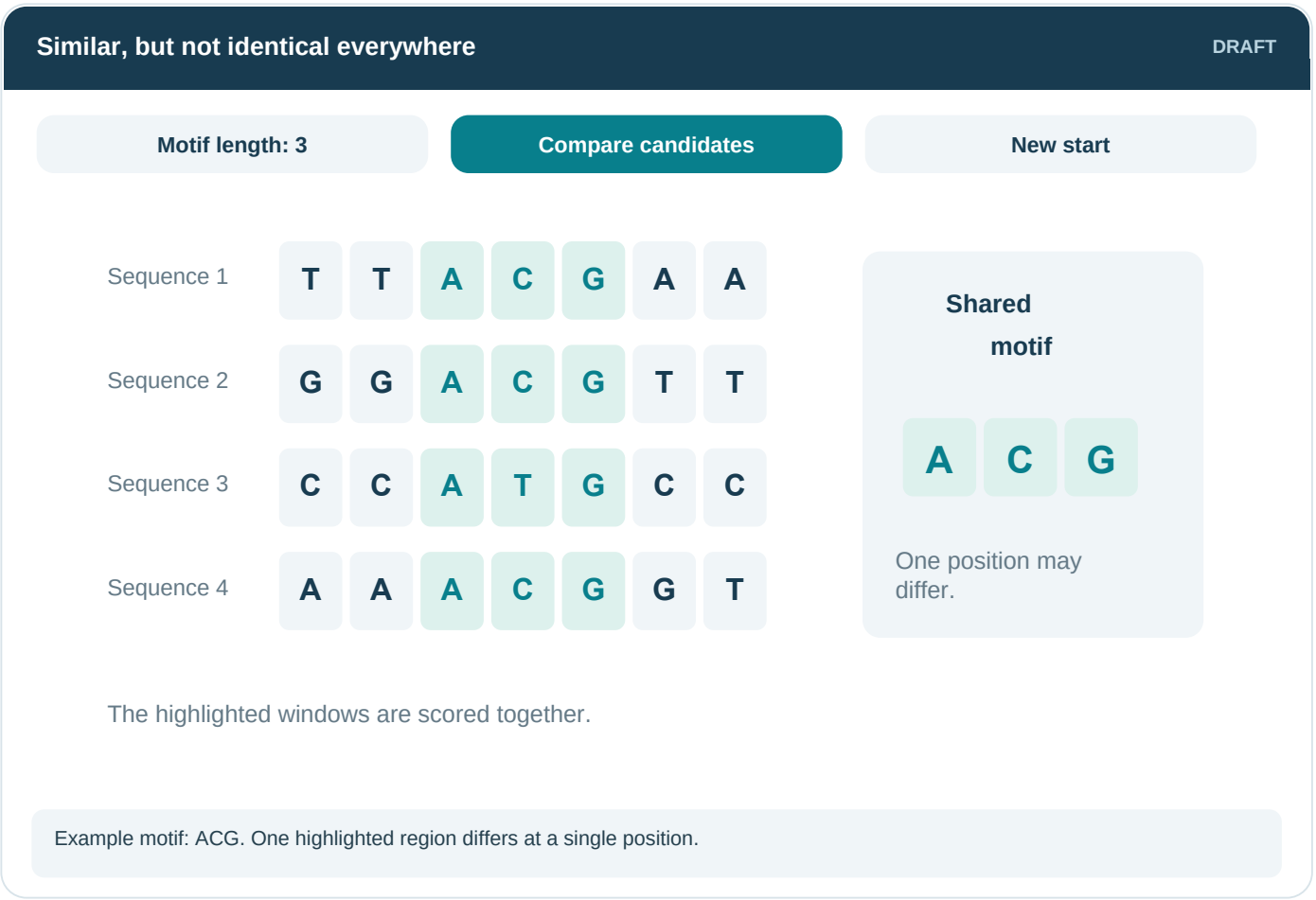
Combine frequencies, strand orientation and base distribution into an informed hypothesis.

The curve alone does not prove a replication origin. Extends the DNA workbench K07.

Find a shared DNA motif

Which short motif occurs in several DNA sequences?

Each sequence may contain a slightly different version of the same motif. Mark candidate regions and compare which selection fits together best.



EXPLORE IT STEP BY STEP

1. Choose sequences
2. Find similar windows
3. Show shared features

WHAT YOU WILL DISCOVER

See why examining several sequences together helps, and why random starts can suggest different candidates.

A similar motif is initially a hypothesis about biological function.

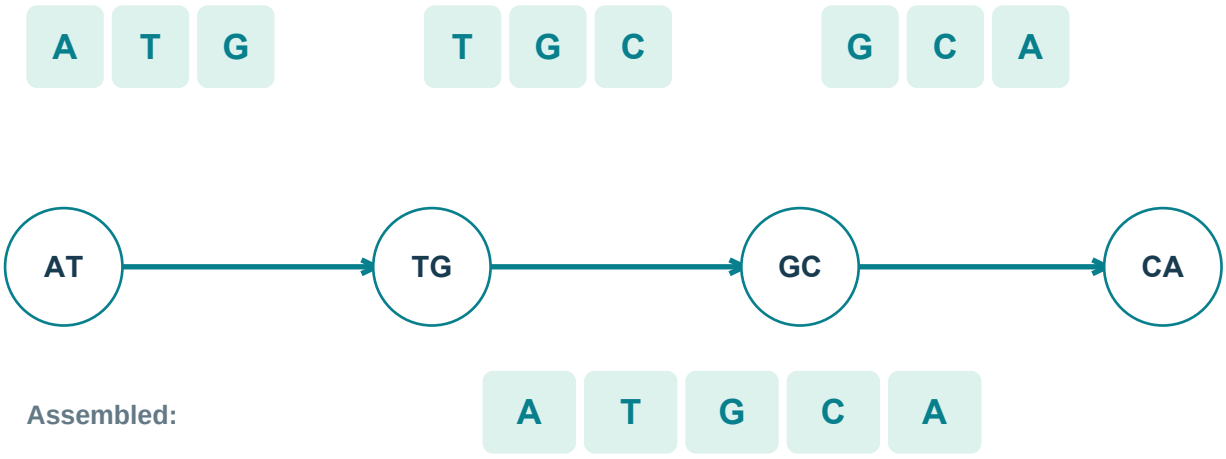
Build a genome from fragments

How can short DNA fragments form a longer sequence?

Arrange overlapping fragments. A connection diagram helps identify matching ends and makes ambiguity visible.

Reassemble overlapping fragments DRAFT

Choose fragments | Show connections | Follow the path



ATG, TGC and GCA can be assembled into ATGCA.

EXPLORE IT STEP BY STEP

1. Compare fragments
2. Join matching ends
3. Assemble a sequence

WHAT YOU WILL DISCOVER

Discover why repeated fragments can allow several possible continuations.

The introduction uses error-free examples; real experimental data are harder to work with.

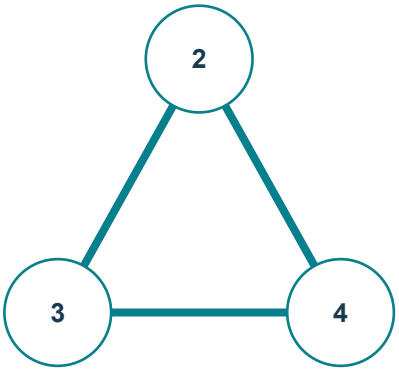
Infer a cyclic peptide from masses

Can fragment masses reveal a molecular ring?

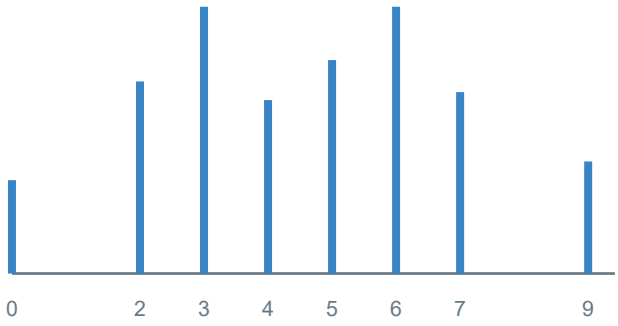
A peptide is a short chain of building blocks, here joined into a ring. Compare possible arrangements with the masses of its fragments.

Which sequence explains the measurement? DRAFT

Example spectrum | Check candidates | Add noise



A candidate ring



Compare these with the measurement.

Simplified example: three building blocks with masses of 2, 3 and 4.

EXPLORE IT STEP BY STEP

1. Inspect masses
2. Build candidate rings
3. Check the match

WHAT YOU WILL DISCOVER

See how unsuitable candidates are eliminated early and how missing measurements make the search harder.

The building-block masses are deliberately simplified teaching values.

Compare two biological sequences

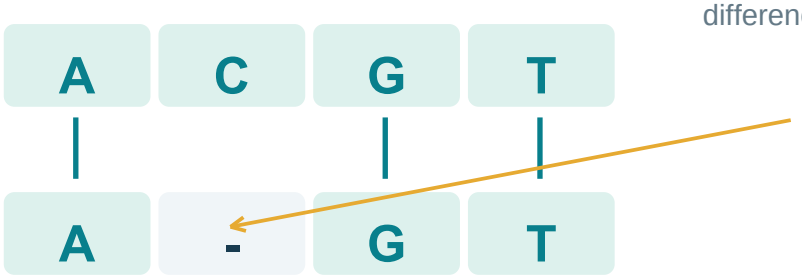
How do sequences align when parts are missing or changed?

Place two DNA or protein sequences one above the other. Allowing gaps reveals which regions match well.

Similarity despite missing letters DRAFT

Choose two sequences | Global | Local | Show alignment

One possible alignment



A gap reveals the difference.

The order of the letters is preserved.

Example: ACGT and AGT align with a gap beneath the C.

EXPLORE IT STEP BY STEP

1. Select sequences
2. Find an alignment
3. Inspect differences

WHAT YOU WILL DISCOVER

Distinguish a comparison of entire sequences from a particularly similar local region.

The result depends on how differences and gaps are scored.

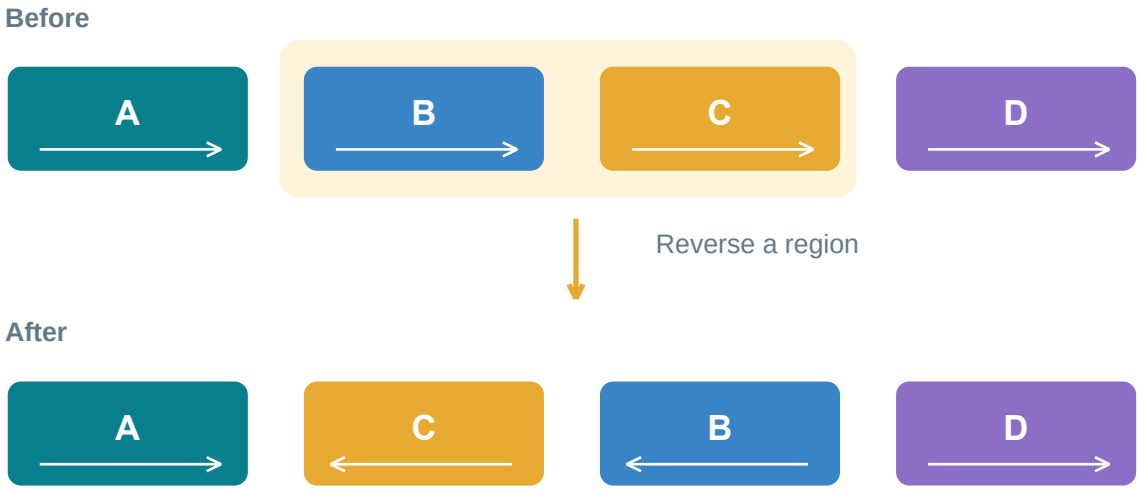
Rearrange large DNA segments

How do genomes differ when segments are reversed?

Represent large DNA segments as labelled blocks. Reverse a region and follow the changes in both order and orientation.

Same building blocks, different order DRAFT

Select a region | Reverse | Compare steps



Reversing a region changes both the order and the orientation of its blocks.

EXPLORE IT STEP BY STEP

1. Compare block orders
2. Reverse a segment
3. Check neighbours

WHAT YOU WILL DISCOVER

See large rearrangements that a simple letter-by-letter comparison struggles to explain.

The models simplify real genomes; a short sequence of moves is not necessarily optimal.

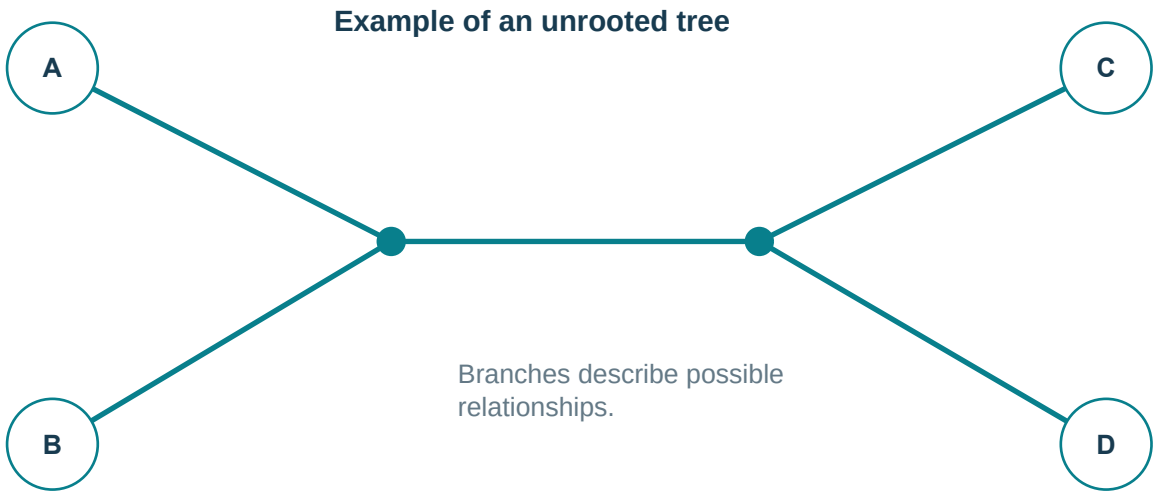
Show relatedness as a tree

Which observed sequences might be more closely related?

Compare differences between examples. Build a tree whose branches describe possible evolutionary relationships.

Differences suggest a tree hypothesis DRAFT

Choose examples | **Build a tree** | Compare methods



The tree topology is a hypothesis derived from the selected data.

EXPLORE IT STEP BY STEP

1. Measure differences
2. Reconstruct a tree
3. Compare assumptions

WHAT YOU WILL DISCOVER

Discover how different assumptions and data can produce different evolutionary trees.

Such a tree does not establish a confirmed history of direct transmission.

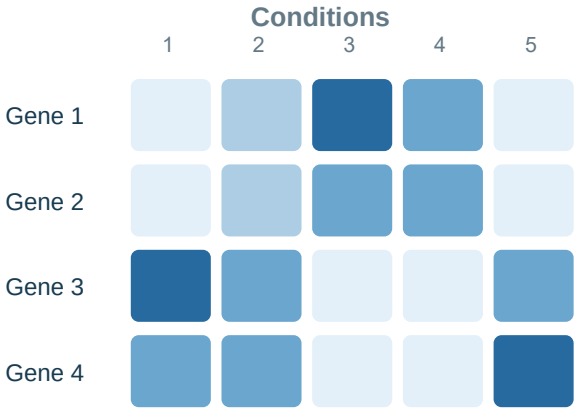
Find genes with similar patterns

Which measurement series rise and fall together?

Colours show how active genes are under different conditions. Group similar patterns and compare different notions of similarity.

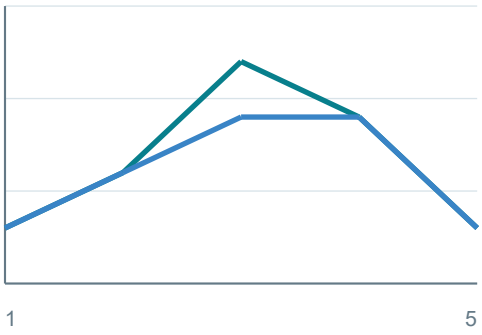
Which genes respond similarly? DRAFT

Choose a data series | **Change scaling** | Compare groups



Light: less active / Dark: more active

Similar patterns



Gene 1 and gene 2

Example: genes 1 and 2 show similar patterns; genes 3 and 4 show a different pattern.

EXPLORE IT STEP BY STEP

1. Inspect data series →
2. Group patterns →
3. Compare views

WHAT YOU WILL DISCOVER

Discover why matching patterns and matching absolute measurements are different things.

Similar patterns do not prove a shared function. Extends K17.

Find DNA fragments in a long sequence

Where might a short DNA fragment have come from?

Search for many short fragments in a longer reference sequence. A prebuilt index narrows down candidate locations for closer comparison.

An indexed reference makes repeated searches easier

DRAFT

Choose a reference | Fragment: ACG | Check matches

Reference sequence

T	T	A	C	G	T	T	A	C	G	A

Query fragment

A	C	G

→ Position 3 | Position 8

Multiple matches remain visible.

ACG matches at two positions in this example, so its origin is ambiguous.

EXPLORE IT STEP BY STEP

1. Prepare a reference
2. Find candidate sites
3. Compare more closely

WHAT YOU WILL DISCOVER

See the benefit of a prebuilt index and the challenge of multiple matching positions.

A detected difference alone does not explain a disease.

Infer hidden states

What might be behind an observed sequence?

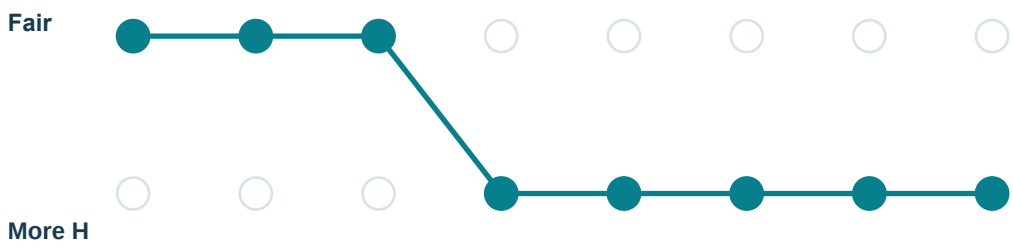
Imagine two coins: one fair and one biased towards heads. You see only the tosses. A model seeks a plausible sequence of the coins used.

You see the outcomes, not their cause DRAFT

Choose tosses | Show inferred states | Inspect uncertainty

H	T	H	H	H	H	T	H

H = heads / T = tails



A plausible switch, not directly observed.

Colours show inferred states. Several explanations may fit the tosses.

EXPLORE IT STEP BY STEP

1. Inspect observations
2. Check possible states
3. Understand uncertainty

WHAT YOU WILL DISCOVER

The same basic principle helps describe changing regions in DNA or characteristic patterns in protein families.

A plausible hidden sequence is not a directly observed certainty.

How convincing is a protein match?

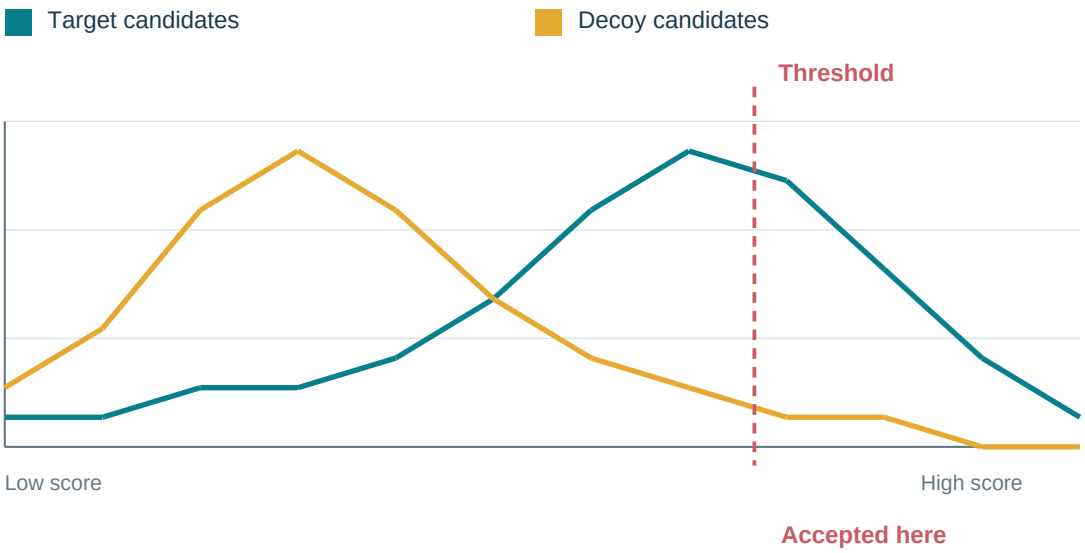
Does a candidate fit well, or is it best just by chance?

Compare measured fragments with candidate peptides. Artificial comparison candidates show how easily high scores can also arise by chance.

A good match needs a baseline

DRAFT

Choose measurement | Compare candidates | Move threshold



The distributions are illustrative. Moving the threshold changes which matches are accepted.

EXPLORE IT STEP BY STEP

1. Score candidates
2. Estimate chance matches
3. Choose a threshold

WHAT YOU WILL DISCOVER

Distinguish a high score from convincing evidence and see how the search-space size affects the result.

An estimated error rate applies to a set of matches, not as certainty about each individual match.